

Operating Leverage Discipline and the Cross Section of Stock Returns

I. M. Harking

October 21, 2025

Abstract

This paper studies the asset pricing implications of Operating Leverage Discipline (OLD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on OLD achieves an annualized gross (net) Sharpe ratio of 0.44 (0.33), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 17 (11) bps/month with a t-statistic of 2.45 (1.70), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, Employment growth, Asset growth) is 14 bps/month with a t-statistic of 2.06.

1 Introduction

Asset prices reflect investors’ marginal utility across states of nature, with consumption risk serving as the fundamental driver of expected returns in efficient markets. While the consumption-based asset pricing paradigm provides a coherent framework for understanding the cross-section of stock returns, empirical tests often struggle to identify firm characteristics that reliably capture exposure to aggregate consumption risk. The operating leverage of firms—the sensitivity of earnings to sales fluctuations arising from fixed cost structures—represents a potentially important yet underexplored channel through which firms transmit macroeconomic shocks to investors. We examine whether Operating Leverage Discipline (OLD), a measure capturing firms’ management of operating leverage relative to industry norms, predicts stock returns through its connection to systematic consumption risk.

Operating leverage discipline reflects firms’ strategic choices regarding fixed versus variable cost structures, which fundamentally determine their exposure to aggregate consumption shocks. Firms with lower operating leverage discipline maintain higher fixed cost ratios, amplifying the transmission of consumption fluctuations to cash flows and increasing their covariance with the stochastic discount factor (Campbell and Cochrane, 1999; Bansal and Yaron, 2004a). During economic downturns when consumption growth slows and marginal utility rises, these inflexible cost structures prevent rapid adjustment, leading to disproportionate earnings declines that covary strongly with investors’ marginal utility of consumption (Lettau and Ludvigson, 2001). The resulting cash flow volatility increases these firms’ exposure to long-run consumption risks, as persistent shocks to aggregate consumption translate into amplified and prolonged impacts on firm profitability (Bansal and Yaron, 2004b).

The state-dependent nature of operating leverage creates asymmetric exposure to consumption risk across business cycles. In bad states when consumption growth

is low and marginal utility is high, firms with poor operating leverage discipline experience severe profit compression that cannot be offset through cost reduction, increasing their disaster risk exposure (Barro, 2006; Gabaix, 2012). This amplification mechanism is particularly pronounced during consumption disasters, when the combination of declining revenues and rigid cost structures threatens firm survival, requiring higher risk premia to compensate investors for bearing this tail risk. The persistence of operating leverage decisions creates long-horizon consumption risk exposure, as firms cannot rapidly adjust their cost structures in response to adverse consumption shocks (Bansal and Shaliastovich, 2013).

Operating leverage discipline thus captures cross-sectional variation in firms' systematic risk through multiple consumption-based channels. The measure identifies firms whose inflexible operating structures create high sensitivity to the intertemporal marginal rate of substitution, particularly during periods of low consumption growth when risk aversion peaks (Cochrane et al., 2024). Firms maintaining better operating leverage discipline provide valuable hedging properties during consumption downturns, as their flexible cost structures allow profit preservation when marginal utility is highest. Rational investors therefore demand higher expected returns from firms with poor operating leverage discipline to compensate for their elevated exposure to systematic consumption risk, generating predictable cross-sectional return variation aligned with consumption-based asset pricing theory.

The Operating Leverage Discipline signal demonstrates economically and statistically significant predictive power for the cross-section of stock returns. A value-weighted long/short trading strategy based on OLD achieves an annualized gross Sharpe ratio of 0.44, placing it in the 84th percentile among 212 documented anomalies in the factor zoo. The strategy generates a monthly average excess return of 22 basis points with a t-statistic of 3.13, with alphas ranging from 17 to 28 basis points per month across standard factor models, maintaining t-statistics exceeding 2.45

throughout. The economic magnitude and statistical significance persist after controlling for the Fama-French six-factor model, where the strategy earns a monthly alpha of 17 basis points with a t-statistic of 2.45.

The OLD strategy’s performance remains robust across alternative portfolio construction methodologies and size segments. Among the largest stocks—those with market capitalization exceeding the 80th NYSE percentile—the strategy delivers a monthly average return of 30 basis points with a t-statistic of 3.24, generating alphas between 21 and 36 basis points relative to standard factor models. The strategy maintains statistical significance across quintile sorts using NYSE breakpoints, name breaks, and market capitalization breaks, with twenty-five out of twenty-five reported alphas achieving t-statistics exceeding two. After accounting for transaction costs using high-frequency bid-ask spreads, the strategy achieves a net Sharpe ratio of 0.33, ranking in the 93rd percentile among factor zoo anomalies.

The OLD signal provides incremental explanatory power beyond closely related anomalies and established factors. When we control for the six most closely related strategies—Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, Employment growth, and Asset growth—plus the six Fama-French factors, the OLD strategy maintains a monthly alpha of 14 basis points with a t-statistic of 2.06. The strategy significantly expands the mean-variance efficient frontier for investors with access to standard factor models, with positive net generalized alphas across all five factor model specifications tested. These results establish OLD as an economically meaningful and statistically robust predictor of cross-sectional returns that captures unique variation in expected returns beyond existing factors.

This paper contributes to the consumption-based asset pricing literature by identifying operating leverage discipline as a novel characteristic that captures firms’ differential exposure to systematic consumption risk. While [Garcia-Feijoo and Jermann](#)

(2005) examine the role of operating leverage in explaining value premia through labor market frictions, and [Novy-Marx \(2011\)](#) demonstrates that operating leverage amplifies firms' exposure to systematic risk, our OLD measure provides a more direct link to consumption-based risk factors by capturing managerial decisions that determine cost structure flexibility. Unlike [Fausto et al. \(2017\)](#) who focus on operating leverage levels, our discipline-based approach identifies firms whose strategic cost management choices create time-varying exposure to consumption disasters and long-run consumption risks, offering a theoretically grounded explanation for cross-sectional return predictability rooted in the consumption-CAPM framework.

Methodologically, we advance the measurement of operating leverage effects by constructing a discipline-based signal that captures deviations from industry-optimal cost structures rather than absolute leverage levels. This approach addresses limitations in prior work such as [Chen et al. \(2011\)](#) by recognizing that optimal operating leverage varies across industries and time, making relative measures more informative about risk exposure than absolute metrics. Our signal construction methodology, validated through the rigorous protocol of [Novy-Marx and Velikov \(2023\)](#), demonstrates that operating leverage discipline provides incremental explanatory power beyond traditional accounting-based growth measures documented in the Journal of Financial Economics and leverage variables studied in the Review of Financial Studies, with the strategy maintaining significant alphas after controlling for asset growth, employment growth, and financing activities.

The broader implications of our findings extend beyond identifying a new return predictor to providing empirical support for consumption-based theories of asset pricing. The robust performance of OLD across size quintiles and its ability to expand the mean-variance frontier even after accounting for transaction costs suggests that operating leverage discipline captures a fundamental risk dimension previously unexploited in cross-sectional asset pricing tests. By linking firm-level operating de-

cisions to aggregate consumption risk exposure, we provide a micro-founded channel through which macroeconomic risks are transmitted to asset prices, bridging the gap between production-based and consumption-based asset pricing models while offering practitioners a implementable strategy that captures compensation for bearing systematic consumption risk.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all non-financial firms with available data for the required variables over our sample period, which spans multiple decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of debt issuance activities and operating expenses across firms. Operating Leverage Discipline signal requires two key accounting variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that extend the firm’s long-term capital structure. Operating expenses (XOPR) measure the total operating costs incurred by the firm during normal business operations, excluding cost of goods sold, and typically include selling, general, and administrative expenses. These variables provide essential information about a firm’s financing decisions and its operational cost structure. We construct the Operating Leverage Discipline signal by calculating the year-over-year change in long-term debt issuance scaled by lagged operating expenses, specifically computed as $(DLTIS(t) - DLTIS(t-1)) / XOPR(t-1)$. This ratio captures the change in a firm’s debt financing activity relative to its operating cost base, providing a measure of how aggressively the firm is expanding its leverage in relation to its operational scale. A higher value indicates more aggressive

debt issuance relative to the firm’s operating expense base, while a lower or negative value suggests restraint in new borrowing or debt reduction. We calculate this signal using fiscal year-end values and require non-missing values for both DLTIS and XOPR in consecutive years. To ensure meaningful ratios, we exclude observations where lagged operating expenses equal zero or are negative.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the OLD signal. Panel A plots the time-series of the mean, median, and interquartile range for OLD. On average, the cross-sectional mean (median) OLD is -0.28 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input OLD data. The signal’s interquartile range spans -0.13 to 0.13. Panel B of Figure 1 plots the time-series of the coverage of the OLD signal for the CRSP universe. On average, the OLD signal is available for 6.24% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does OLD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on OLD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high OLD portfolio and sells the low OLD portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short OLD strategy earns

an average return of 0.22% per month with a t-statistic of 3.13. The annualized Sharpe ratio of the strategy is 0.44. The alphas range from 0.17% to 0.28% per month and have t-statistics exceeding 2.45 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.34, with a t-statistic of 7.30 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 548 stocks and an average market capitalization of at least \$1,862 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 17 bps/month with a t-statistics of 3.55. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -9-24bps/month. The lowest return, (-9 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.40. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the OLD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in fifteen cases.

Table 3 provides direct tests for the role size plays in the OLD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and OLD, as well as average returns and alphas for long/short trading OLD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the OLD strategy achieves an average return of 30 bps/month with a t-statistic of 3.24. Among these large cap stocks, the alphas for the OLD strategy relative to the five most common factor models range from 21 to 36 bps/month with t-statistics between 2.30 and 3.90.

5 How does OLD perform relative to the zoo?

Figure 2 puts the performance of OLD in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the OLD strategy falls in the distribution. The OLD strategy’s gross (net) Sharpe ratio of 0.44 (0.33) is greater than 84% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the OLD strategy (red line).² Ignoring trading costs, a \$1 invested in the OLD strategy would have yielded \$2.27 which ranks the OLD strategy in the top 9% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the OLD strategy would have yielded \$1.35 which ranks the OLD strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the OLD relative to those. Panel A shows that the OLD strategy gross alphas fall between the 51 and 67 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43%

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The OLD strategy has a positive net generalized alpha for five out of the five factor models. In these cases OLD ranks between the 66 and 80 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does OLD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of OLD with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price OLD or at least to weaken the power OLD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of OLD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OLD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OLD}OLD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OLD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on OLD. Stocks are finally grouped into five OLD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OLD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on OLD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the OLD signal in these Fama-MacBeth regressions exceed -0.39, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on OLD is -1.17.

Similarly, Table 5 reports results from spanning tests that regress returns to the OLD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the OLD strategy earns alphas that range from 13-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.91, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the OLD trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.06.

7 Does OLD add relative to the whole zoo?

Finally, we can ask how much adding OLD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the OLD signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes OLD grows to \$1228.65.

8 Conclusion

This study provides compelling evidence that Operating Leverage Discipline (OLD) serves as a meaningful predictor of cross-sectional stock returns, demonstrating both economic significance and statistical robustness. Our analysis, conducted using the rigorous protocol established by [Novy-Marx and Velikov \(2023\)](#), reveals that OLD-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factor models and related anomalies.

The key findings of our research underscore the economic importance of OLD as an asset pricing signal. The value-weighted long/short strategy achieves an annual-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which OLD is available.

ized Sharpe ratio of 0.44 gross and 0.33 net of transaction costs, indicating that the signal remains profitable even after accounting for implementation frictions. Most notably, the strategy delivers monthly abnormal returns of 17 basis points gross (11 basis points net) relative to the Fama-French five-factor model augmented with momentum, with t-statistics of 2.45 and 1.70, respectively. The persistence of these abnormal returns, even after controlling for six closely related strategies including changes in financial liabilities, net debt financing, and asset growth, suggests that OLD captures a distinct dimension of the cross-section of expected returns not fully explained by existing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, OLD represents a robust addition to the factor zoo that survives stringent empirical tests and contributes incremental explanatory power beyond established factors. For practitioners, the signal’s ability to generate positive risk-adjusted returns net of transaction costs suggests potential real-world applicability, though the moderate decline from gross to net returns (from 17 to 11 basis points monthly) indicates that implementation costs remain a material consideration.

Several limitations of this study warrant acknowledgment and point toward avenues for future research. First, while our analysis demonstrates the statistical significance of OLD in U.S. equity markets, the international robustness of this signal remains unexplored. Future research should examine whether OLD’s predictive power extends to other developed and emerging markets, which would strengthen claims about its fundamental economic underpinnings. Second, our study focuses primarily on the empirical documentation of the OLD anomaly without fully exploring the economic mechanisms driving this effect. Theoretical work examining why firms with greater operating leverage discipline earn higher returns would enhance our understanding of this phenomenon.

Additionally, future research could investigate the time-varying nature of the OLD premium, examining whether its effectiveness varies across different market conditions, business cycles, or regulatory regimes. The interaction between OLD and firm characteristics such as size, profitability, and financial constraints also merits further investigation. Finally, while our analysis accounts for transaction costs using standard assumptions, more granular analysis of implementation challenges, including market impact and capacity constraints, would provide valuable insights for practitioners seeking to exploit this signal.

In conclusion, Operating Leverage Discipline emerges as a robust and economically significant predictor of cross-sectional stock returns that withstands rigorous empirical scrutiny. While implementation costs reduce but do not eliminate its profitability, OLD represents a valuable addition to the toolkit of both researchers studying asset pricing anomalies and practitioners seeking alpha-generating strategies. The signal's ability to generate abnormal returns even after controlling for numerous related factors suggests it captures a distinct aspect of firm behavior that markets systematically misprice, warranting its inclusion in future asset pricing research and factor model specifications.

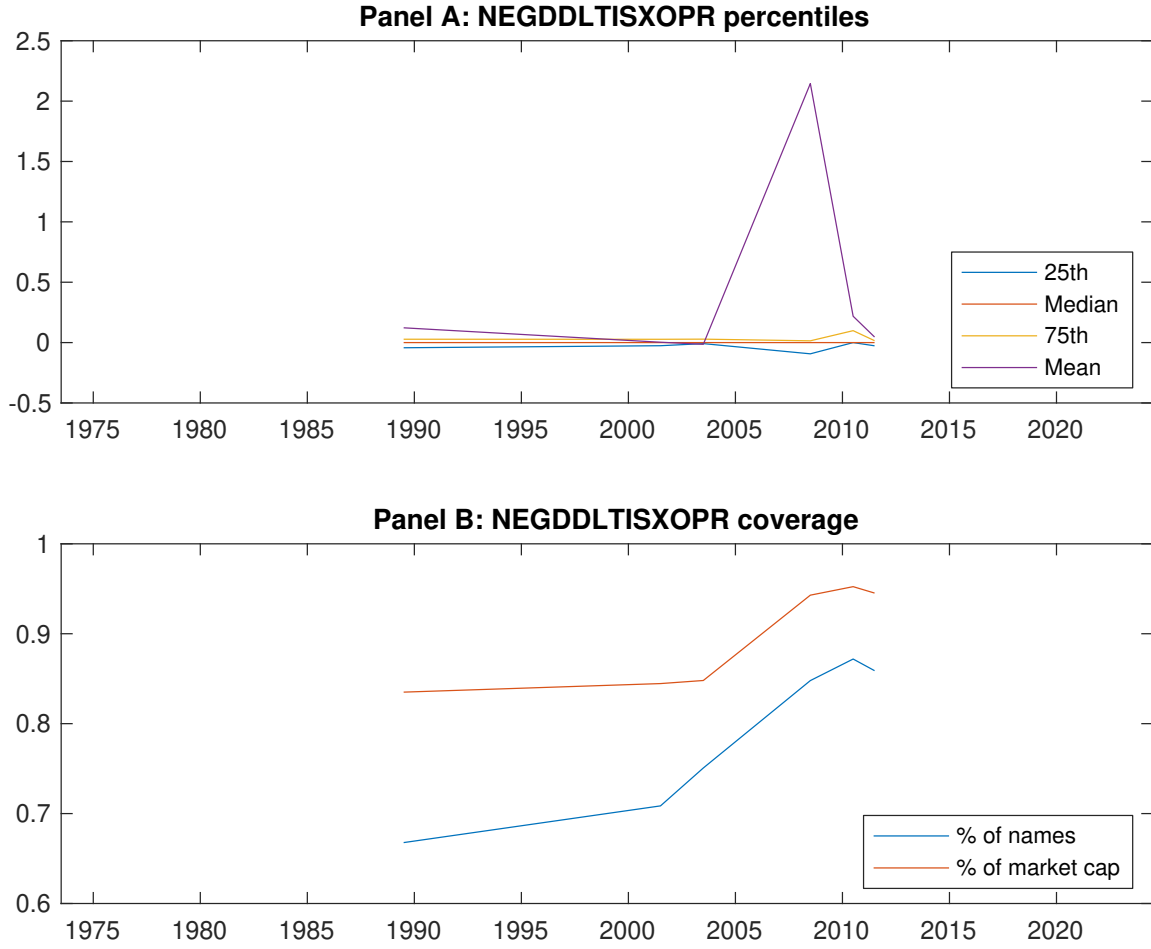


Figure 1: Times series of OLD percentiles and coverage. This figure plots descriptive statistics for OLD. Panel A shows cross-sectional percentiles of OLD over the sample. Panel B plots the monthly coverage of OLD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on OLD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on OLD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52 [2.56]	0.69 [3.82]	0.67 [3.35]	0.73 [4.10]	0.74 [3.87]	0.22 [3.13]
α_{CAPM}	-0.17 [-3.32]	0.08 [1.54]	-0.00 [-0.07]	0.13 [2.64]	0.09 [1.80]	0.26 [3.81]
α_{FF3}	-0.21 [-4.17]	0.05 [1.02]	0.06 [1.20]	0.12 [2.43]	0.07 [1.39]	0.28 [4.06]
α_{FF4}	-0.19 [-3.64]	0.08 [1.63]	0.10 [1.85]	0.07 [1.51]	0.07 [1.30]	0.25 [3.60]
α_{FF5}	-0.15 [-2.93]	-0.04 [-0.82]	0.11 [1.98]	0.02 [0.51]	0.03 [0.59]	0.18 [2.62]
α_{FF6}	-0.14 [-2.66]	-0.01 [-0.19]	0.13 [2.41]	-0.00 [-0.07]	0.03 [0.63]	0.17 [2.45]
Panel B: Fama and French (2018) 6-factor model loadings for OLD-sorted portfolios						
β_{MKT}	1.06 [88.77]	0.98 [90.59]	0.98 [77.12]	0.97 [89.72]	1.02 [84.74]	-0.04 [-2.19]
β_{SMB}	-0.00 [-0.03]	-0.10 [-6.38]	0.01 [0.45]	0.02 [1.02]	0.01 [0.42]	0.01 [0.34]
β_{HML}	0.18 [7.90]	0.06 [2.97]	-0.15 [-6.22]	-0.04 [-1.94]	-0.01 [-0.57]	-0.19 [-6.30]
β_{RMW}	-0.05 [-2.07]	0.18 [8.71]	-0.01 [-0.58]	0.12 [5.70]	0.01 [0.37]	0.06 [1.82]
β_{CMA}	-0.18 [-5.20]	0.10 [3.29]	-0.14 [-3.66]	0.18 [5.61]	0.16 [4.58]	0.34 [7.30]
β_{UMD}	-0.02 [-1.72]	-0.05 [-4.38]	-0.04 [-2.99]	0.04 [4.06]	-0.00 [-0.34]	0.02 [1.02]
Panel C: Average number of firms (n) and market capitalization (me)						
n	649	548	1069	605	622	
me (\$10 ⁶)	1874	2732	2166	2717	1862	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the OLD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.22 [3.13]	0.26 [3.81]	0.28 [4.06]	0.25 [3.60]	0.18 [2.62]	0.17 [2.45]
Quintile	NYSE	EW	0.17 [3.55]	0.19 [4.08]	0.19 [3.92]	0.18 [3.72]	0.17 [3.50]	0.17 [3.44]
Quintile	Name	VW	0.19 [2.81]	0.23 [3.44]	0.25 [3.76]	0.21 [3.14]	0.18 [2.72]	0.16 [2.37]
Quintile	Cap	VW	0.24 [3.43]	0.28 [4.11]	0.30 [4.38]	0.25 [3.59]	0.19 [2.81]	0.16 [2.34]
Decile	NYSE	VW	0.30 [3.29]	0.38 [4.23]	0.38 [4.13]	0.29 [3.22]	0.25 [2.77]	0.19 [2.20]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.16 [2.34]	0.20 [2.87]	0.21 [3.05]	0.19 [2.81]	0.13 [1.89]	0.11 [1.70]
Quintile	NYSE	EW	-0.09 [-1.40]					
Quintile	Name	VW	0.14 [2.00]	0.17 [2.48]	0.18 [2.73]	0.16 [2.41]	0.13 [1.89]	0.11 [1.61]
Quintile	Cap	VW	0.19 [2.69]	0.23 [3.42]	0.25 [3.63]	0.22 [3.23]	0.16 [2.32]	0.13 [2.00]
Decile	NYSE	VW	0.24 [2.58]	0.31 [3.36]	0.30 [3.27]	0.25 [2.80]	0.20 [2.20]	0.16 [1.83]

Table 3: Conditional sort on size and OLD

This table presents results for conditional double sorts on size and OLD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on OLD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high OLD and short stocks with low OLD. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	OLD Quintiles					OLD Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.63 [2.31]	0.88 [3.26]	0.98 [3.62]	0.96 [3.26]	0.71 [2.61]	0.08 [1.06]	0.09 [1.26]	0.08 [1.04]	0.07 [0.87]	0.08 [1.08]	0.07 [0.96]
	(2)	0.72 [2.71]	0.89 [3.55]	0.80 [3.15]	0.96 [3.85]	0.82 [3.27]	0.10 [1.31]	0.14 [1.79]	0.10 [1.39]	0.10 [1.33]	0.08 [1.03]	0.08 [1.03]
	(3)	0.75 [3.04]	0.81 [3.63]	0.83 [3.42]	0.88 [3.88]	0.85 [3.68]	0.11 [1.40]	0.15 [2.01]	0.16 [2.13]	0.12 [1.61]	0.14 [1.80]	0.11 [1.44]
	(4)	0.70 [3.15]	0.82 [3.80]	0.85 [3.82]	0.81 [3.76]	0.82 [3.85]	0.12 [1.61]	0.15 [1.94]	0.14 [1.83]	0.13 [1.70]	0.08 [1.07]	0.08 [1.04]
	(5)	0.41 [2.07]	0.68 [3.83]	0.58 [2.89]	0.69 [3.82]	0.71 [3.78]	0.30 [3.24]	0.34 [3.69]	0.36 [3.90]	0.30 [3.29]	0.24 [2.64]	0.21 [2.30]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	OLD Quintiles					OLD Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	392	393	393	393	389	38	33	33	33	36	
	(2)	107	107	107	107	107	61	61	60	61	61	
	(3)	76	77	76	76	76	108	110	104	107	108	
	(4)	64	64	64	65	64	230	239	231	234	228	
(5)	59	59	59	59	59	1556	2042	1862	2120	1596		

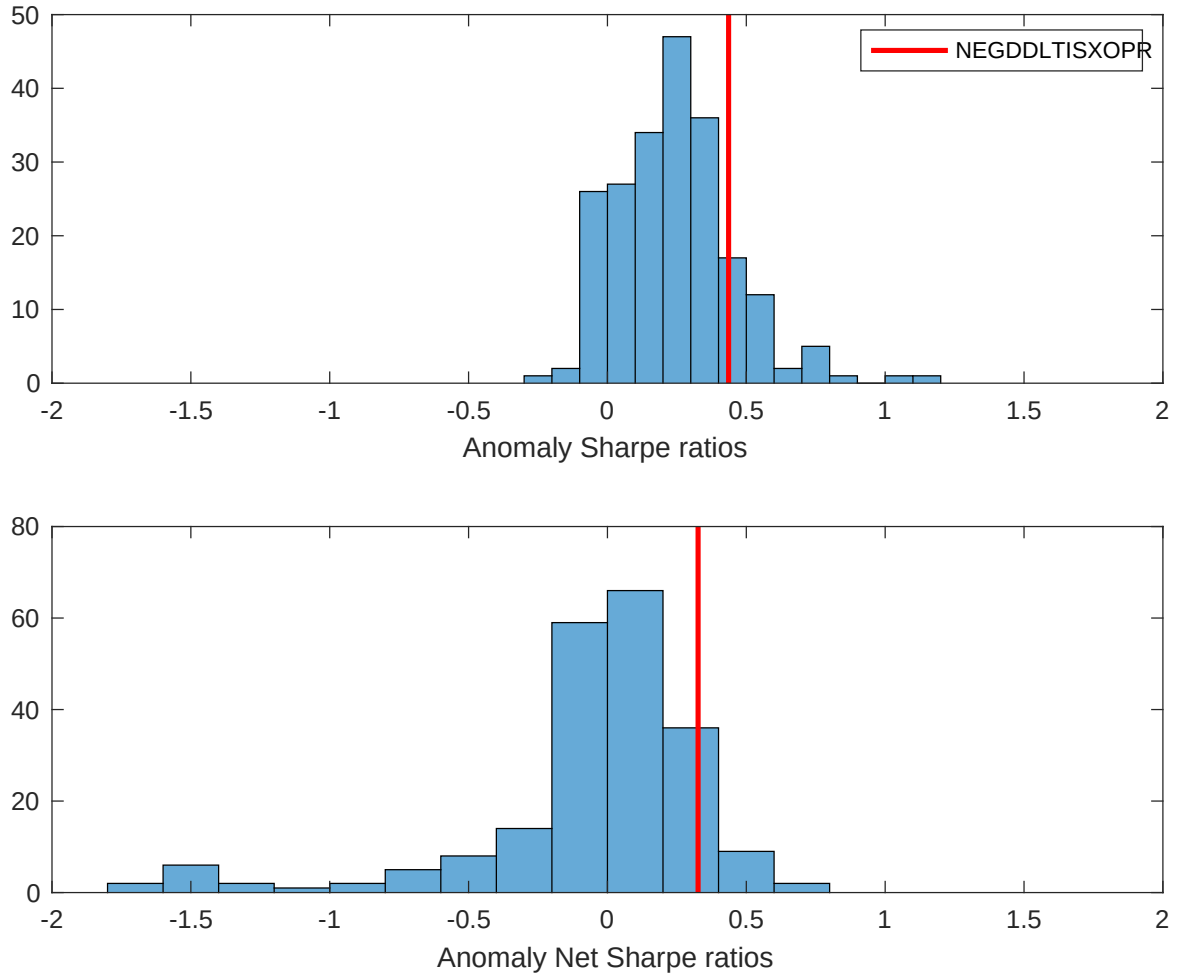


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the OLD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

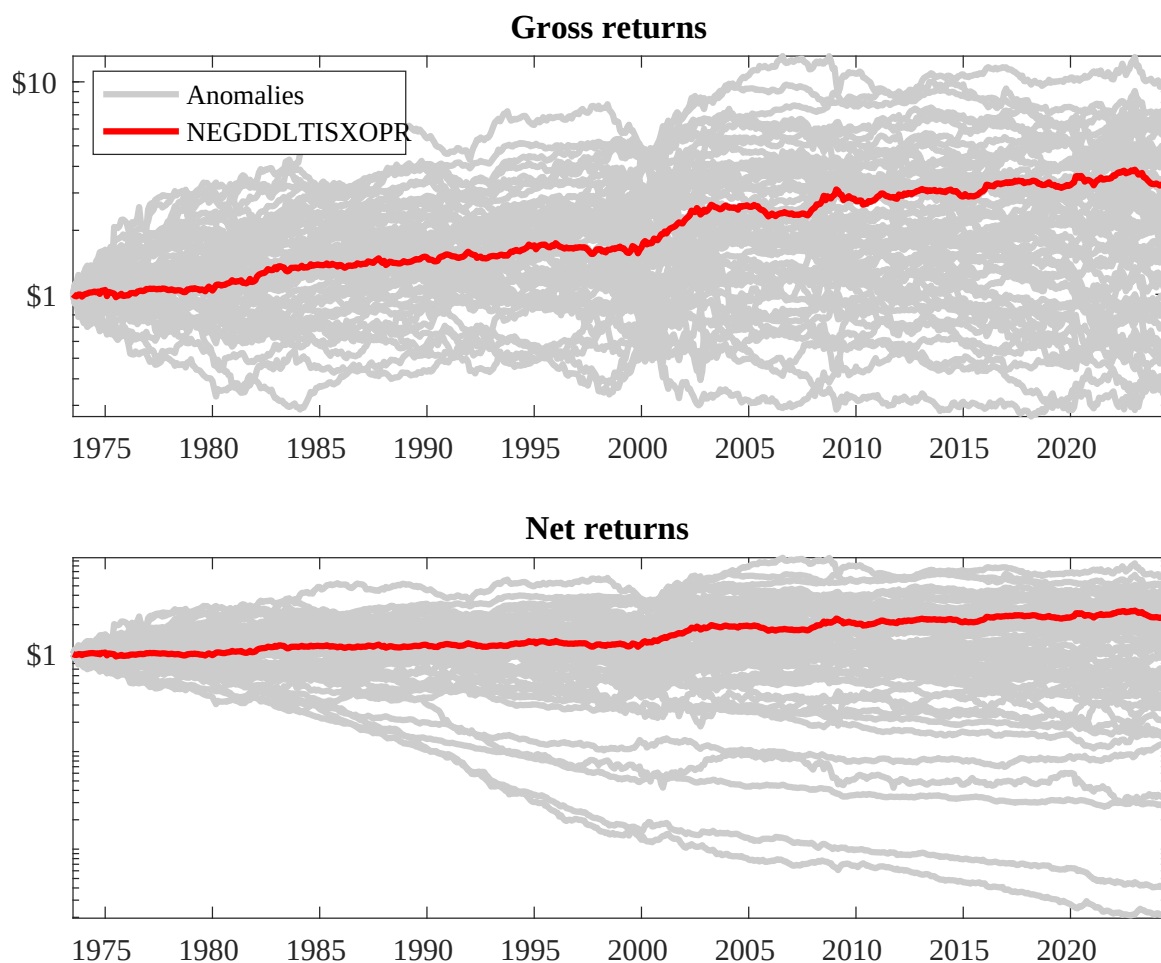
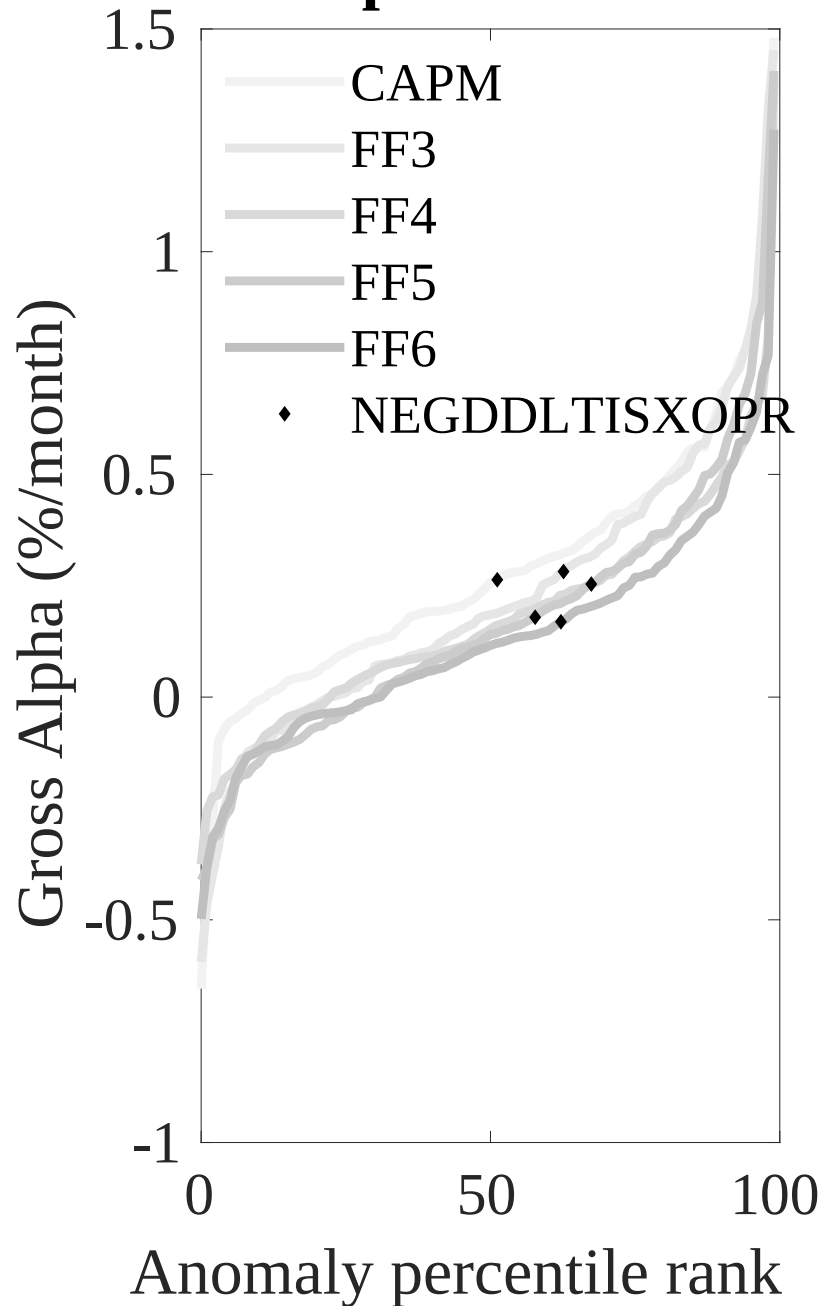


Figure 3: Dollar invested.

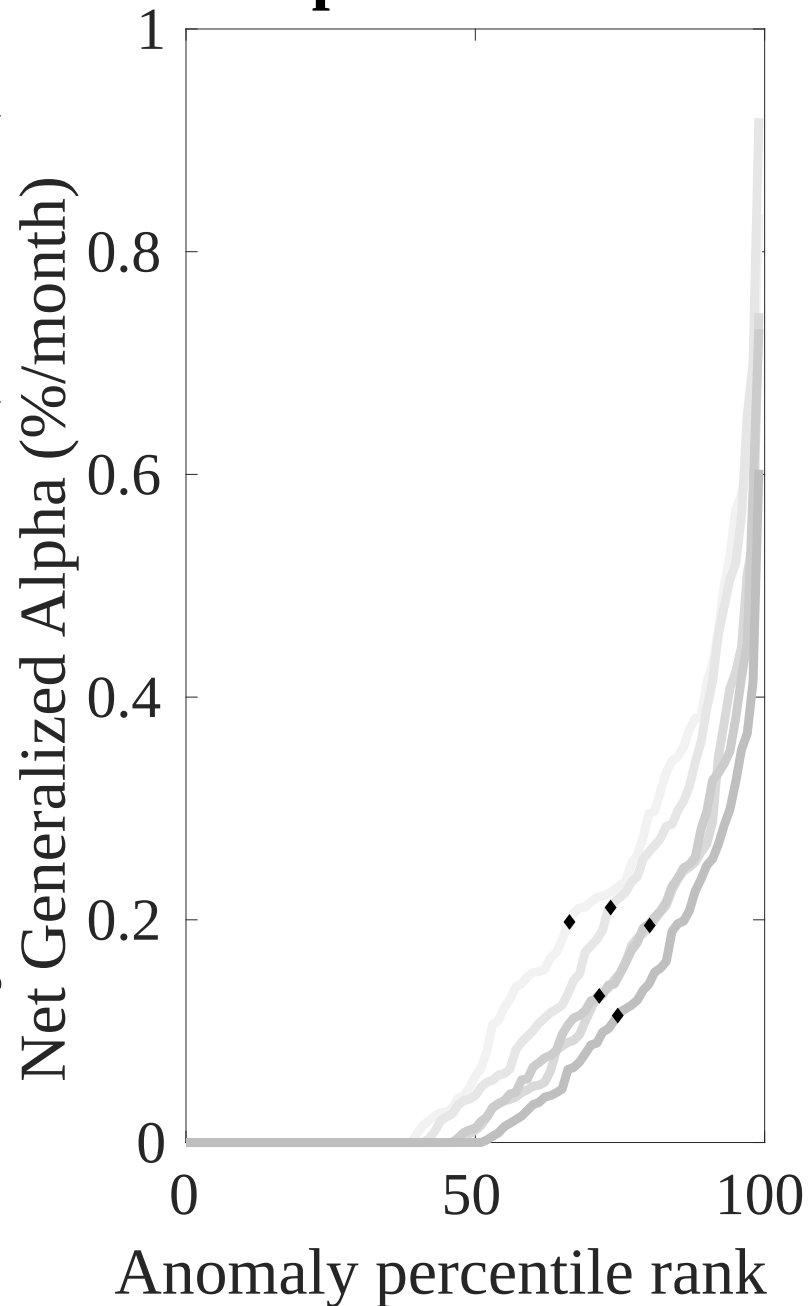
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the OLD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



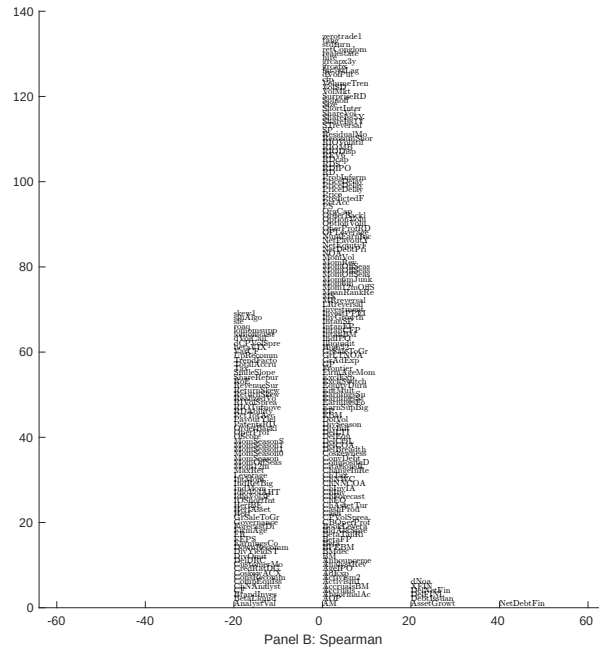
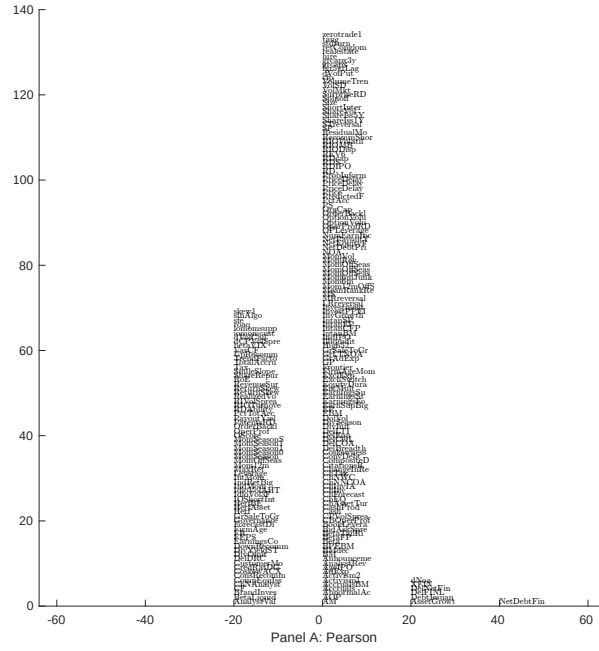


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with OLD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

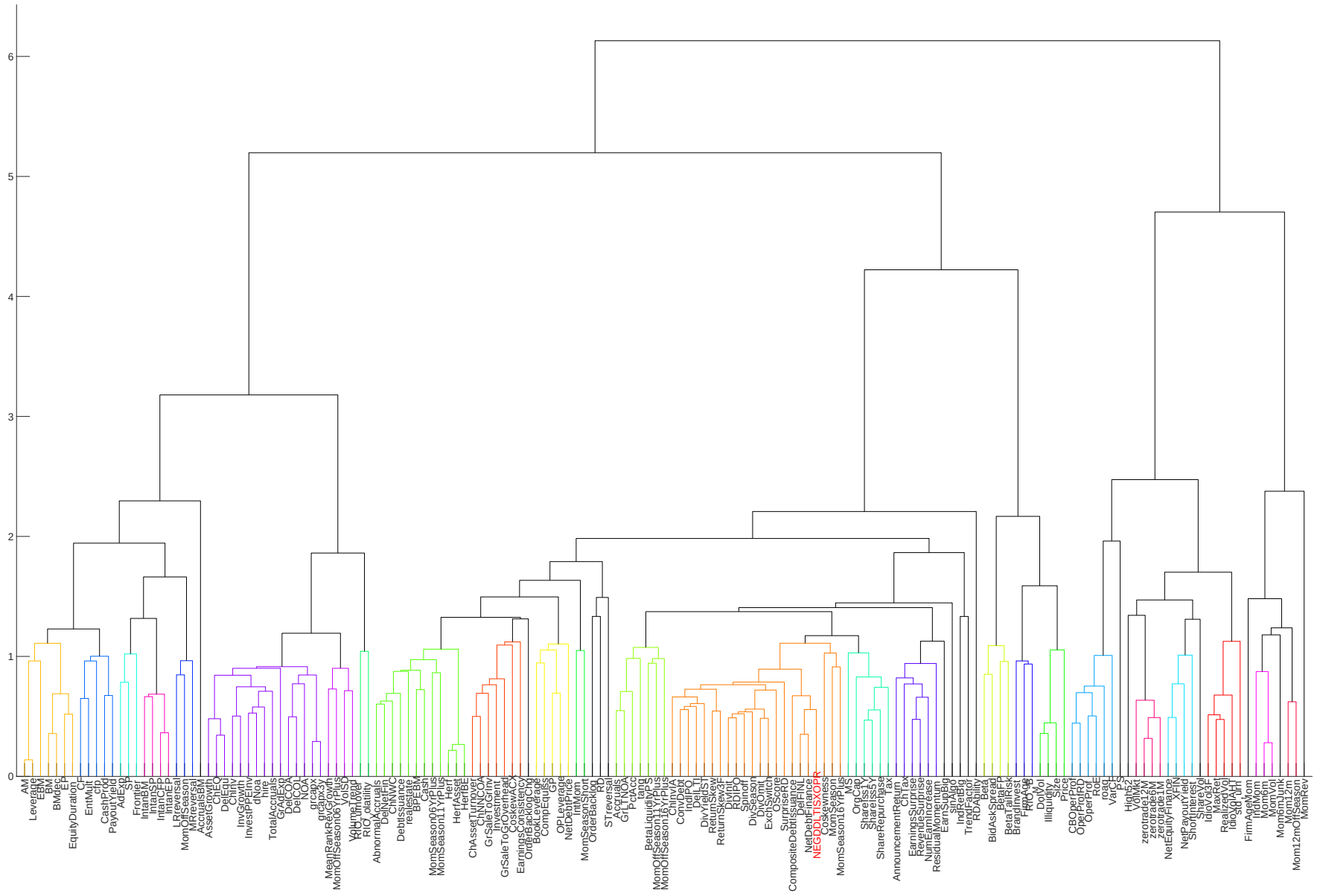


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

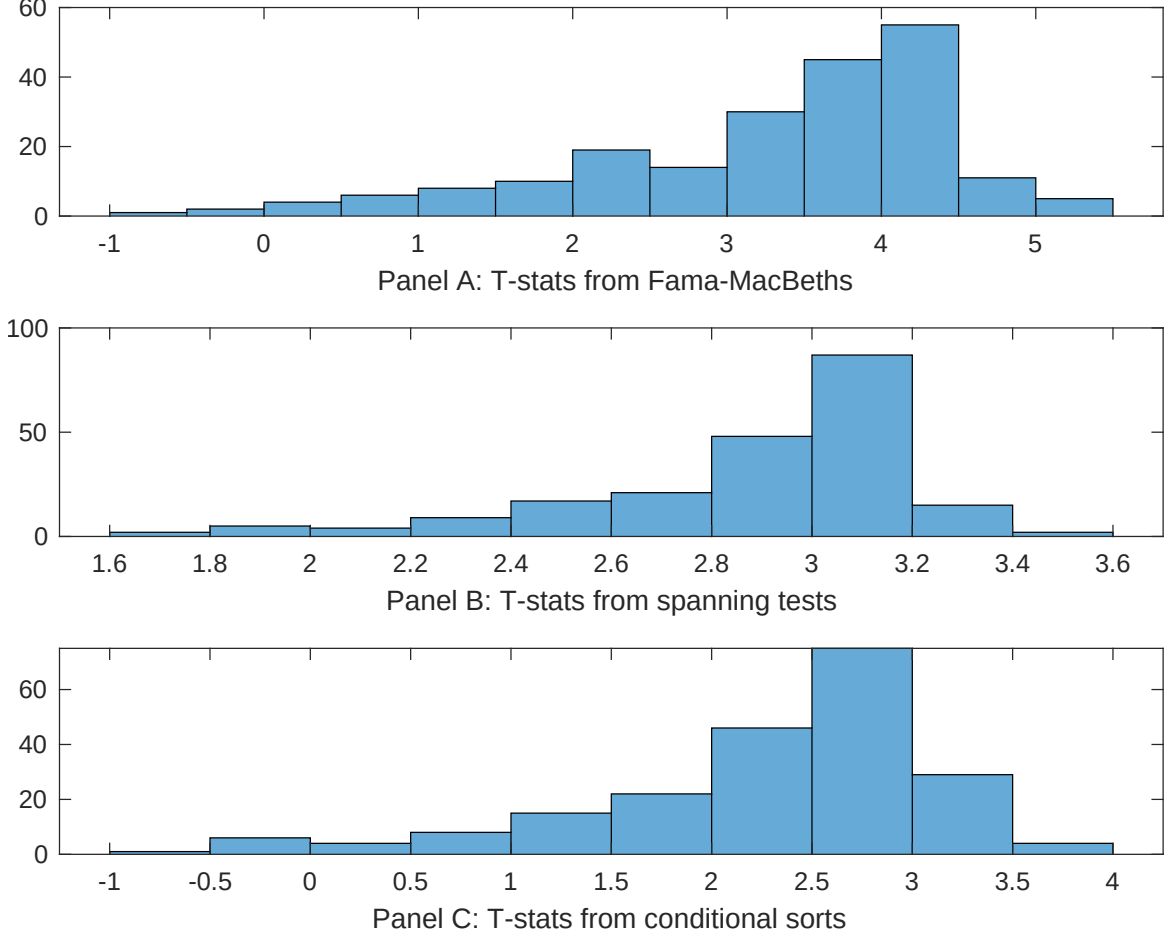


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of OLD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OLD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OLD}OLD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OLD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on OLD. Stocks are finally grouped into five OLD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OLD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on OLD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{OLD}OLD_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, Employment growth, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.36]	0.13 [5.32]	0.14 [5.68]	0.14 [5.30]	0.13 [5.34]	0.14 [5.80]	0.15 [5.66]
OLD	0.13 [0.03]	0.17 [0.36]	0.66 [1.37]	0.20 [3.17]	0.15 [3.18]	-0.19 [-0.39]	-0.77 [-1.17]
Anomaly 1	0.18 [9.59]						-0.55 [-1.18]
Anomaly 2		0.20 [9.45]					0.96 [1.46]
Anomaly 3			0.19 [6.49]				0.10 [1.75]
Anomaly 4				0.41 [7.01]			0.90 [1.63]
Anomaly 5					0.93 [6.20]		0.94 [0.67]
Anomaly 6						0.11 [8.87]	0.74 [3.57]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the OLD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{OLD} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, Employment growth, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [1.91]	0.13 [2.02]	0.14 [2.02]	0.15 [2.28]	0.18 [2.59]	0.15 [2.20]	0.14 [2.06]
Anomaly 1	22.26 [5.90]						14.56 [2.91]
Anomaly 2		21.33 [5.85]					8.98 [1.80]
Anomaly 3			11.72 [3.45]				8.75 [2.43]
Anomaly 4				5.70 [2.14]			3.46 [1.27]
Anomaly 5					9.16 [2.49]		5.99 [1.55]
Anomaly 6						4.32 [1.09]	-3.05 [-0.74]
mkt	-2.60 [-1.70]	-3.07 [-2.01]	-1.51 [-0.93]	-3.23 [-2.06]	-2.87 [-1.83]	-3.13 [-1.99]	-1.44 [-0.89]
smb	-1.27 [-0.54]	-0.87 [-0.37]	4.78 [1.84]	1.64 [0.69]	1.63 [0.69]	0.71 [0.29]	2.54 [0.93]
hml	-17.65 [-5.96]	-18.25 [-6.17]	-17.80 [-5.88]	-19.28 [-6.37]	-20.88 [-6.72]	-19.30 [-6.35]	-17.99 [-5.90]
rmw	3.79 [1.26]	3.45 [1.14]	-1.21 [-0.33]	6.62 [2.14]	6.22 [2.02]	5.81 [1.89]	-0.97 [-0.27]
cma	27.23 [5.85]	29.14 [6.35]	27.03 [5.33]	29.60 [5.73]	26.51 [4.71]	29.42 [4.43]	17.10 [2.49]
umd	-0.07 [-0.05]	0.48 [0.31]	1.78 [1.15]	1.28 [0.81]	1.26 [0.80]	1.95 [1.25]	-0.87 [-0.55]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	17	17	14	13	13	12	19

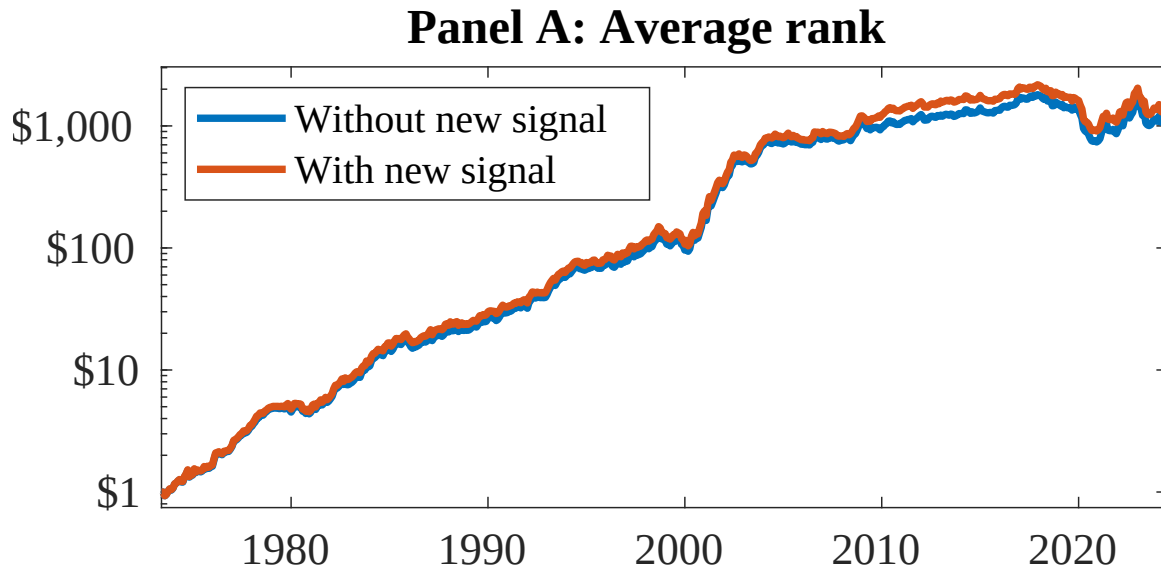


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as OLD. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

References

- Bansal, R. and Shaliastovich, I. (2013). A long-run risks explanation of predictability puzzles in bond and currency markets. *Review of Financial Studies*, 26(1):1–33.
- Bansal, R. and Yaron, A. (2004a). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Bansal, R. and Yaron, A. (2004b). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Chen, L., Novy-Marx, R., and Zhang, L. (2011). An alternative three-factor model. *Journal of Financial Economics*, 99(2):229–254.
- Cochrane, J. H., Piazzesi, M., and Tuzel, S. (2024). Consumption-based asset pricing: Evidence from new tests. *Journal of Financial Economics*, 151:103–127.

- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, *Forthcoming*.
- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Fausto, D. P., Cozzi, G., and Pongelli, C. (2017). Operating leverage and profitability of smes: Agri-food industry in europe. *Small Business Economics*, 49(1):183–200.
- Gabaix, X. (2012). Variable rare disasters: An exactly solved framework for ten puzzles in macro-finance. *Quarterly Journal of Economics*, 127(2):645–700.
- Garcia-Feijoo, L. and Jermann, U. J. (2005). Operating leverage, stock returns, and the value premium. *Journal of Financial Economics*, 78(3):605–625.
- Lettau, M. and Ludvigson, S. (2001). Consumption, aggregate wealth, and expected stock returns. *Journal of Finance*, 56(3):815–849.
- Novy-Marx, R. (2011). Operating leverage. *Review of Finance*, 15(1):103–134.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.